i.a.

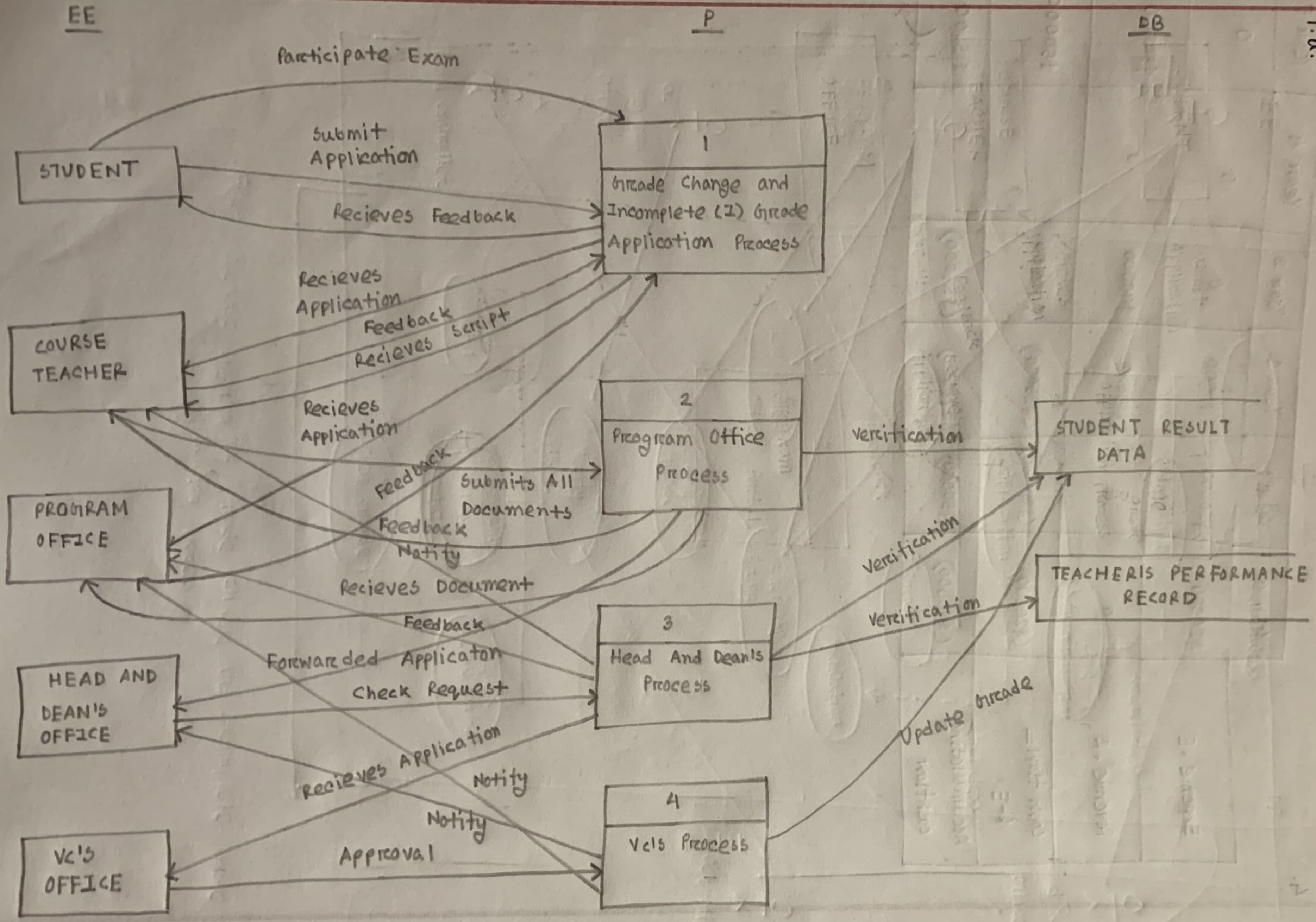


Fig. - DFD of Grade Change and Incomplete Grade Application process b.

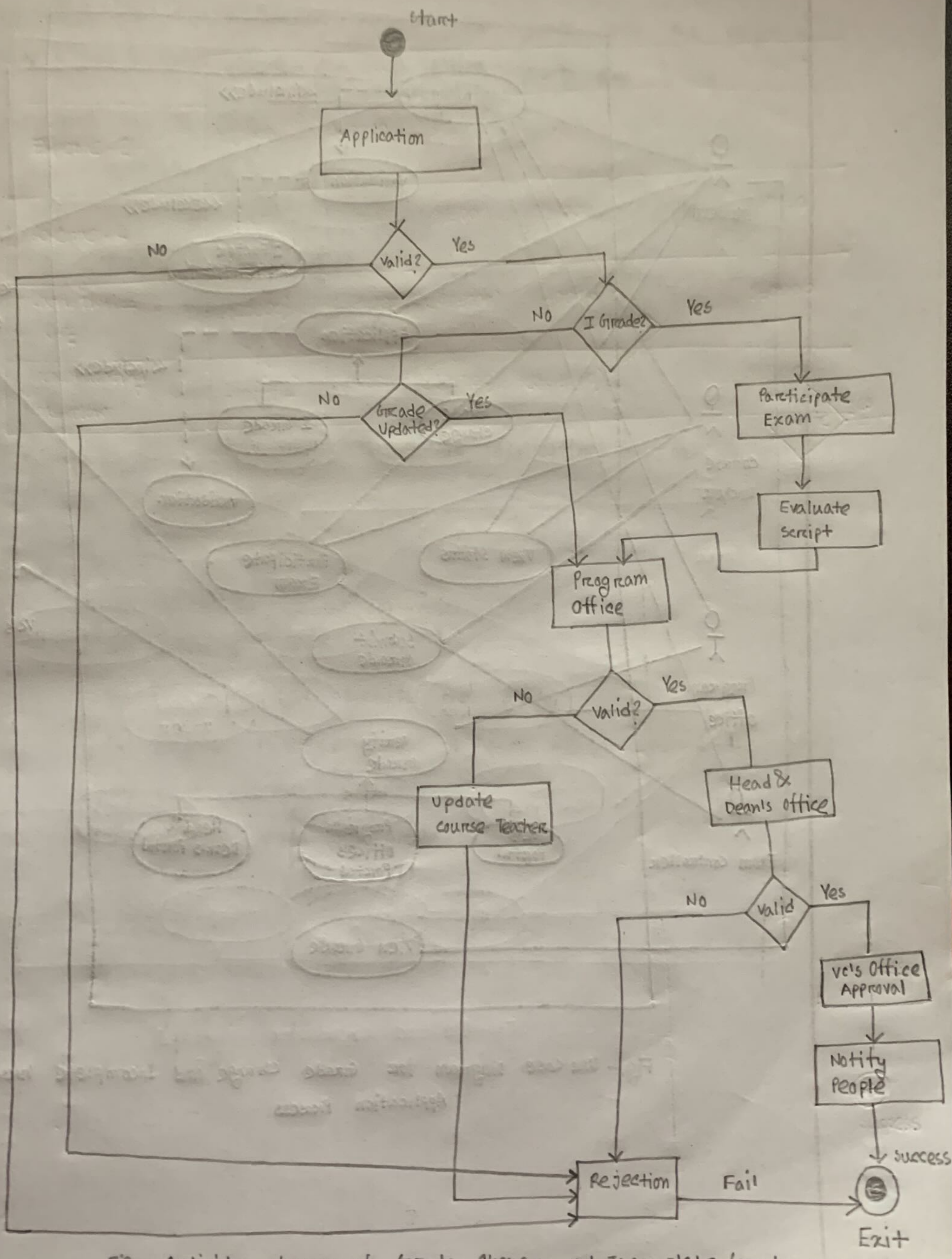


Fig.- Activity Diagram of Grade Change and Incomplete Grade Application Process c.

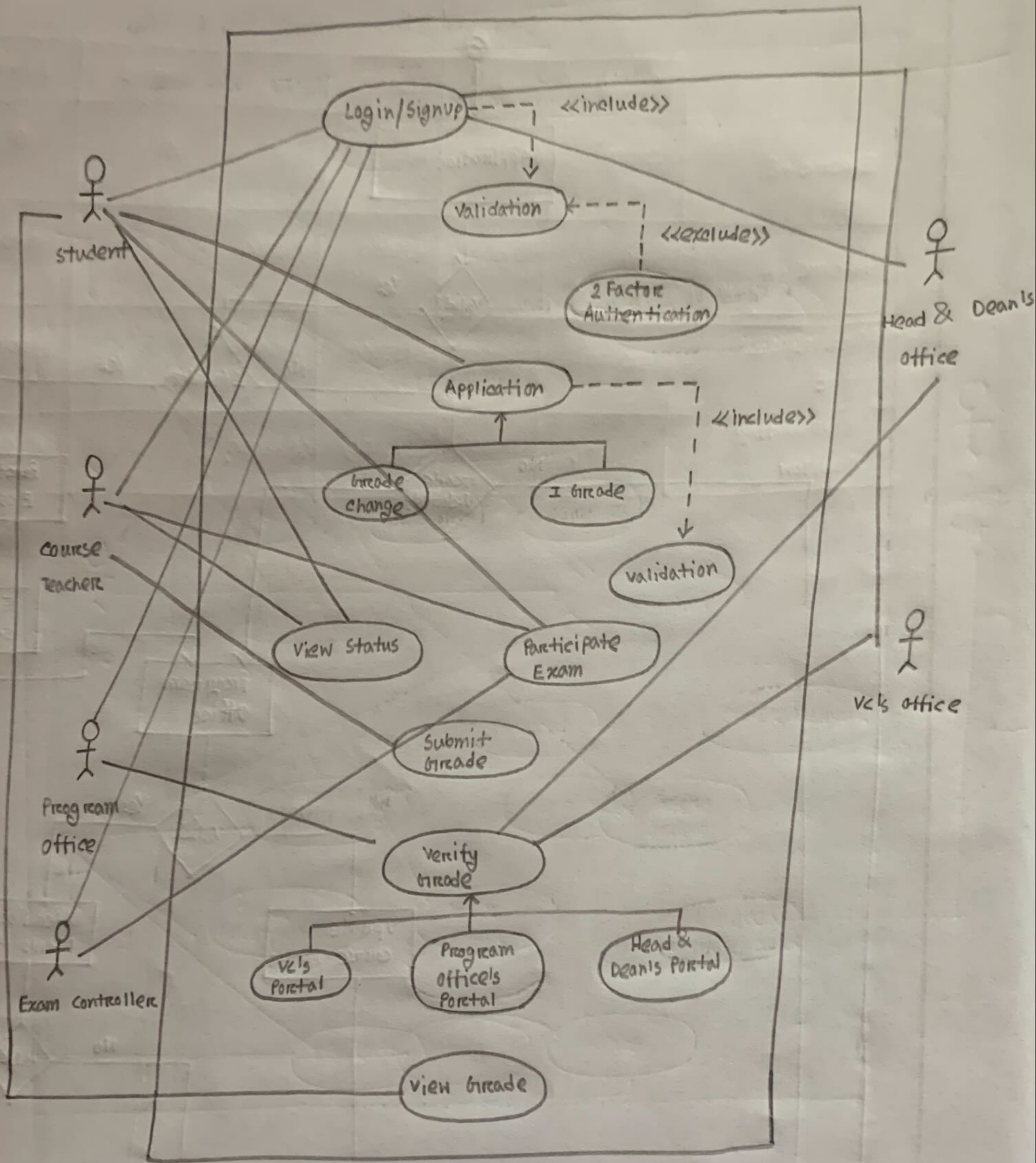


Fig.- Use case Diagram for Grade Change and Incomplete Grade Application Process

d. UC-07 : Verify Grade

Description: The course teacher submits the grade and the program office, Head & Dean's office and VC's office verify the grade

Stakeholders and Interests:

1. Course Teacher
2. Program office
3. Head & Dean's Office
4. VC's office

Primary Actor: Course Teacher

Preconditions: The student must need to do application for the following process

Success Scenario:

1. The student submits the application.
2. The authority verifies it.
3. The course teacher submits the grade.
4. The program office and Head & Dean's office approve the grade.
5. VC's final approval.
6. The student views his grade.

Alternative Scenario:

1. The student submits the application
2. Authority rejects it.
3. Student views his status.

Post-conditions: Student gets his CGPA updated.

2. The online (surveys) would be most beneficial in gathering information for this project.

| Merits | Demerits |
| --- | --- |
| 1. Less skill to administer & economical | 1. Require knowledge for possible answers |
| 2. Can target mass audience. | 2. No option to clarify doubts on both sides. |

Other two less suitable tools can be on-site observation & JAD. In on-site observation there might be some lacks present in bottlenecks which are beyond discoverable and in JAD we can't ensure the presence of all stake holders at a time. Hence, our choice is better.

3. a. No, the DS and ESS aren't work in the same way.

| DSS | ES |
|---|---|
| 1. Make decision by analyzing large amount of datas using models. | 1. AI based system using human base expert system. |
| 2. Doesn't make final decision. | 2. All decisions are final here. |

b. Phases of SDLC:

1. Idea generation
2. Planning
3. Requirement Gathering
4. System Analysis
5. Design
6. Development
7. Testing
8. Deployment
9. Post-maintainence

Requirement gathering is an important SDLC step which ensures the software development process based on a clear, concise and comprehensive understanding of the customer's needs and requirements. It is essential because sometimes the end product doesn't comply with the customers need.

4.

| | Year 0 | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---|---|---|---|---|---|---|
| Expense, E | $20000 | $30000 | $40000 | 0 | 0 | 0 |
| Revenue, R | 0 | 0 | $15000 | $40000 | 0 | $50000 |
| Cash flow = R-E | ($20000) | ($30000) | ($25000) | $40000 | | $50000 |
| Accumulating cashflow | ($20000) | ($50000) | ($75000) | ($35000) | | $15000 |

Profit